

Exam 3 prep

Note: $X \geq 30$ min $X \leq 30$ max

1. $f_y(y; \theta) = \begin{cases} e^{-(y-\theta)} & 0 \leq y \leq \infty \\ 0 & \text{o.w.} \end{cases}$ show $y_{\min}$ is sufficient for θ

Sufficient

$$L(\theta) = \prod_{i=1}^n e^{-(y_i - \theta)} \rightarrow e^{-\sum_{i=1}^n (y_i - \theta)} \cdot I_{[0, \infty)}(y_{\min})$$

$$= e^{-\sum_{i=1}^n y_i} \cdot e^{n\theta} \cdot I_{[0, \infty)}(y_{\min})$$

$\hat{\theta} = y_{\min}$ is

$$h(\hat{\theta}; \theta) \cdot g(y_1, \dots, y_n) \text{ sufficient for } \theta \text{ by factorization Thm 2}$$

$$(e^{n\theta} \cdot I_{[0, \infty)}(y_{\min})) \cdot e^{-\sum_{i=1}^n y_i}$$

Uniform

2. Is $2\bar{X}$ consistent for β $f_x(x) = \begin{cases} 1/\beta & 0 < x \leq \beta \\ 0 & \text{o.w.} \end{cases}$

First

show

unbiasedness

$$E(2\bar{X}) = 2E(\bar{X}) \Rightarrow 2 \cdot \beta/2 = \beta$$

$$\text{Then } \text{Var}(X) = \beta^2/12 \quad \lim_{n \rightarrow \infty} P(|\bar{X}_n - \beta| < \epsilon)$$

$$\geq \lim_{n \rightarrow \infty} 1 - \frac{\text{Var}(2\bar{X}_n)}{\epsilon^2} \Rightarrow \lim_{n \rightarrow \infty} 1 - \frac{4 \text{Var}(\bar{X}_n)}{\epsilon^2}$$

Consistent

$$\text{Var}(\bar{X}) = \text{Var}(\bar{X}_n) \quad 1 - \frac{4 \text{Var}(\bar{X})}{\epsilon^2} \Rightarrow 1 - \frac{4 \cdot \beta^2/12}{n\epsilon^2}$$

$$\Rightarrow 1 - \frac{\beta^2}{3n\epsilon^2} \quad \lim_{n \rightarrow \infty} 1 - \frac{\beta^2}{3n\epsilon^2} = 1 - 0 = 1 \text{ for all } \epsilon > 0$$

$$\lim_{n \rightarrow \infty} P(|2\bar{X}_n - \beta|) = 1 \quad \therefore 2\bar{X} \text{ is consistent for } \beta$$

3. $f(x) = p(1-p)^{x-1}$ Show $\bar{x}$ is consistent for p

Geo
metric

$$E(\bar{x}) = 1/p \text{ unbiased}$$

$$\text{Var}(\bar{x}) = \frac{(p-1)}{p^2}$$

$$\lim_{n \rightarrow \infty} P(|\bar{x}_n - p| < \epsilon) \geq \lim_{n \rightarrow \infty} 1 - \frac{\text{var}(\bar{x}_n)}{\epsilon^2}$$

$$\text{Var}(\bar{x}_n) = \frac{\text{var}(\bar{x})}{n} = \frac{(1-p)}{np^2}$$

$$\lim_{n \rightarrow \infty} 1 - \frac{\text{var}(\bar{x}_n)}{\epsilon^2} = \lim_{n \rightarrow \infty} 1 - \frac{(1-p)}{n\epsilon^2 p^2} = 1 - 0 = 1$$

$$\lim_{n \rightarrow \infty} P(|\bar{x}_n - p| < \epsilon) = 1 \therefore \bar{x} \text{ is consistent for } p$$

4. $g_w(w; \theta) = e^{k(w)P(\theta) + S(w) + Q(\theta)}$ Show $\hat{\theta} = \sum_{i=1}^n k(w_i)$ is sufficient for θ

$$L(\theta) = \prod_{i=1}^n e^{k(w_i)P(\theta) + S(w_i) + Q(\theta)} \Rightarrow e^{P(\theta) \sum_{i=1}^n k(w_i)} \cdot e^{\sum_{i=1}^n S(w_i)} \cdot e^{nQ(\theta)}$$

$$h(\hat{\theta}; \theta) \cdot g(w_1, \dots, w_n) = (e^{P(\theta) \sum_{i=1}^n k(w_i)} \cdot e^{nQ(\theta)}) \cdot (e^{\sum_{i=1}^n S(w_i)})$$

$\hat{\theta}$ is sufficient for θ by fact thm 2

5. $f(x) = \begin{cases} \frac{(\theta-1)^{x-1}}{\theta^x} & x=1, 2, \dots, \theta > 0 \\ 0 & \text{o.w.} \end{cases}$ Find a sufficient estimator for θ

$$\prod_{i=1}^n \frac{(\theta-1)^{x_i-1}}{\theta^{x_i}} \Rightarrow \frac{(\theta-1)^{\sum_{i=1}^n x_i - n}}{\theta^{\sum_{i=1}^n x_i}} \Rightarrow \frac{1}{\theta^{\sum_{i=1}^n x_i}} \cdot (\theta-1)^{\sum_{i=1}^n (x_i - 1)}$$

$$(\theta-1)^{\sum_{i=1}^n x_i} \cdot (\theta-1)^n \cdot \frac{1}{\theta^{\sum_{i=1}^n x_i}} \quad h(\hat{\theta}; \theta) \cdot g(x_1, \dots, x_n)$$

$$\hat{\theta} = \sum_{i=1}^n x_i \text{ is sufficient: } \left((\theta-1)^{\sum_{i=1}^n x_i} \cdot (\theta-1)^n \cdot \frac{1}{\theta^{\sum_{i=1}^n x_i}} \right) \cdot (1)$$

for θ by fact thm 2

6. $f(x) = \frac{1}{2\theta^3} x^2 \cdot e^{-x/\theta}$ find sufficient estimator

Sufficient

$$\prod_{i=1}^n \frac{1}{2\theta^3} x_i^2 \cdot e^{-x_i/\theta}$$

$$\left(\frac{1}{2}\right)^n \cdot \left(\frac{1}{\theta^3}\right)^n \cdot \prod_{i=1}^n x_i^2 \cdot e^{-1/\theta \sum x_i}$$

$\sum x_i$ is a sufficient estimator for θ by FT2

$$h(\theta; \theta) \cdot g(x_1, \dots, x_n)$$

$$\left(\left(\frac{1}{2\theta^3}\right)^n \cdot e^{-1/\theta \cdot \sum x_i}\right) \cdot \left(\prod_{i=1}^n x_i^2\right)$$

7. $f(x) = \frac{1}{2} e^{-x/2}$ is $\bar{X}$ sufficient for λ

Sufficient

$$L(\lambda) = \prod_{i=1}^n \frac{1}{2} e^{-x_i/2} \Rightarrow \frac{1}{2^n} e^{-\sum x_i/2}$$

$\bar{X} = \frac{\sum x_i}{n}$
 $\sum x_i = \bar{X}n$

$$\Rightarrow \frac{1}{2^n} e^{-\bar{X}n/2} \Rightarrow h(\bar{X}; \lambda) \cdot g(x_1, \dots, x_n)$$

$$\left(\frac{1}{2^n} e^{-\bar{X}n/2}\right) \cdot (1)$$

By FT2 $\bar{X}$ is sufficient for λ

8. $f(x) = \frac{1}{2} e^{-x/2}$ is $\bar{X}$ consistent for λ

$E(\bar{X}) = \lambda$ unbiased $\checkmark$ $\text{var}(\bar{X}) = \lambda^2$

$\lim_{n \rightarrow \infty} P(|\bar{X}_n - \lambda| < \epsilon) \geq \lim_{n \rightarrow \infty} 1 - \frac{\text{var}(\bar{X}_n)}{\epsilon^2}$

$\text{var}(\bar{X}_n) = \text{var}(\bar{X})/n$ $\lim_{n \rightarrow \infty} 1 - \frac{\text{var}(\bar{X})}{n \epsilon^2}$

$\lim_{n \rightarrow \infty} 1 - \frac{\lambda^2}{n \epsilon^2} = 1 - 0 = 1$ $\lim_{n \rightarrow \infty} P(|\bar{X}_n - \lambda| < \epsilon) = 1$

$\therefore \bar{X}$ is consistent for λ

9. $f(x) = \frac{1}{2\theta^3} x^2 e^{-x/\theta}$ Show $\bar{X}/3$ is consistent for θ

$E(\bar{X}/3) = \frac{1}{3} E(\bar{X}) = \frac{1}{3} \cdot 3\theta = \theta$ unbiased $\checkmark$

$\text{var}(\bar{X}/3) = \theta^2$ $\lim_{n \rightarrow \infty} P\left(\left|\frac{\bar{X}_n}{3} - \theta\right| < \epsilon\right) \leq \lim_{n \rightarrow \infty} 1 - \frac{\text{var}(\bar{X}_n/3)}{\epsilon^2}$

$\Rightarrow \text{var}(\bar{X}_n/3) = \frac{1}{9} \frac{\text{var}(\bar{X})}{n} = \frac{1}{9} \frac{\theta^2}{n} \Rightarrow \lim_{n \rightarrow \infty} 1 - \frac{\theta^2}{9n\epsilon^2} = 1 - 0$

$= 1$ $P\left(\left|\frac{\bar{X}_n}{3} - \theta\right| < \epsilon\right) = 1 \therefore \bar{X}/3$ is consistent for θ